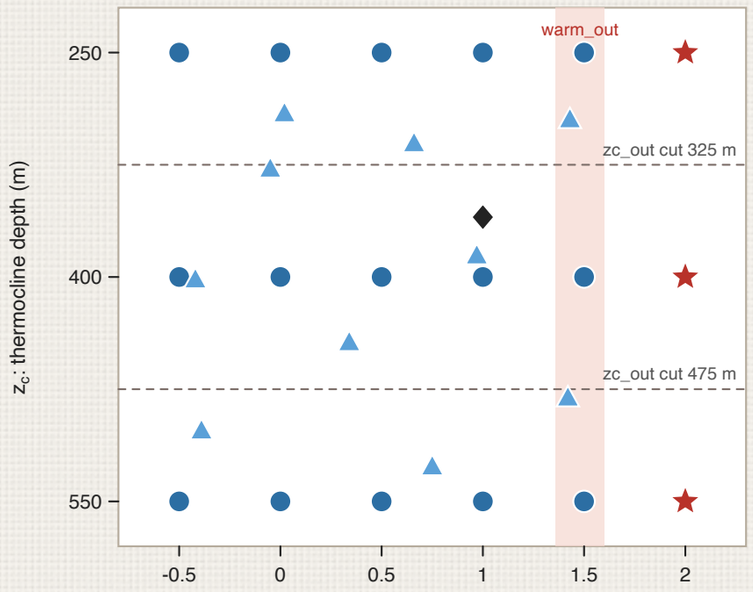
# Experiment 3: can a melt emulator trained on 25 cavity runs extrapolate?

## 1 The runs: 29 MITgcm simulations (Exp. 2)




- 15 factorial runs, s = 1 (training)
- 10 geometry runs, s = 0.8, 1.2 (training)
- 3 final-test runs, $T_{bot}$ = 2.0 (held out)
- Ocean0 reference, linear profile (profile test)

s scales the ice draft; 4 km ISOMIP+ cavity; melt = mean of months 19–24

## 2 What each method is tested on

Each fit sees only its training runs; standardisation, EOFs, kernels and K, n are refitted inside each fold.

**loro** — 25 folds: train 24 / test 1
in range: leave one run out

**zc_out** — 3 folds: thirds of $z_c$
train 17/16/17, test 8/9/8

**warm_out** — 1 fold: train $T_{bot} \le 1.0$ (20)
test $T_{bot} \ge 1.4$ (5)

**profile_test** — train 25 / test Ocean0
different profile; per-cell methods

**final** — train 25 / test $T_{bot}$ = 2.0 (3)
evaluated once, frozen config

training runs  held-out runs

Sensitivity repeat: all folds but final without the 5 runs failing N1 (never used for the hypotheses).

## 3 Eight ways to predict the melt map

**Baseline**

**M0** mean of the training maps

**Physics: one fitted constant**

**M1** $\gamma\, c\, TF$ — Favier et al. 2019, Eq. 2

**M2** $K\, c^2\, |TF|\, TF$ — Burgard et al. 2022, Eq. 14

**M3** $K\, c^2\, TF\, |\langle TF \rangle|$ — Jourdain et al. 2020, Eq. 1

**Statistics on ($T_{bot}$, $z_c$, s)**

**M4** per-cell regression, degree 1 or 2

**M5** Gaussian process on EOFs — Myren & Lawrence 2021

**Neural networks, 5 members**

**M6** deep ensemble on 10 cell features — Lakshminarayanan et al. 2017

**M7** hybrid: $K\, TF_+^n\, g_\theta$(geometry) — $g_\theta$ sees no temperature

$c = \rho_{sw} c_p / (\rho_i L_i)$; TF: thermal forcing at the ice base

## 4 How the predictions are scored (fixed before any fit on real data)

**Metrics, per held-out run (Burgard et al. 2022):**

- RMSE of the melt map over the run's own ice shelf (Eq. 32)
- relative error of the integrated melt (Eq. 33–34)
- error of the mean melt near the grounding line (Eq. 35)
- spread of the 5 network members (M5–M7: also coverage, CRPS)
- noise floor: no difference below 0.157 m $a^{-1}$ or 1.3 % is read

**Pre-registered hypotheses:**

**H3a** On the 3 final-test runs, the hybrid M7 has a smaller integrated-melt error than M6, on average and in at least 2 of the 3 runs (a difference below 1.3 % is inconclusive).

**H3b** Inside the training range (loro), the hybrid costs little: RMSE of M7 ≤ 1.10 × that of M6.

**H3c** Out of range (zc_out, warm_out, final: 33 cases), the ensemble is never confidently wrong: no case with spread < 5 % and integrated error > 20 % (Ockenden et al. 2026 replication).

The ranking of all 8 methods is reported as found, whichever wins.

A schematic of the protocol: no model output or result is drawn in this figure.